

Thanushree Sridhara

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SUMMARY

I am an MSc student in Computer and System Engineering at TU Ilmenau with a strong interest in LLM-based systems. My work focuses on embeddings, semantic search, prompt engineering, and evaluating language model outputs. I'm particularly interested in turning unstructured text into reliable, interpretable information. I'm seeking a Master's thesis or Internship roles starting in September 2026 where I can apply research-driven experimentation to real-world text analysis and information extraction.

SKILLS

- **Programming & Scripting:** Python, SQL
- **AI & Language Processing:** Natural Language Processing (NLP), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Embeddings, BGE Embeddings, Semantic Search, Re-ranking techniques.
- **Software Development & APIs:** FastAPI, Flask, Model Context Protocol (MCP), Git, GitHub
- **Testing & Analysis:** Debugging, Code Testing (pytest)
- **Data & Infrastructure:** Vector Databases (Qdrant, FAISS, ChromaDB), Docker, Kubernetes (AKS), Terraform, Data Ingestion

PROJECTS

Intelligent Document Q&A Platform

- Designed and deployed a production **RAG system** that answers natural language queries over user-uploaded documents with full source attribution and real-time streaming
- Implemented **RAGAS based evaluation pipeline** measuring faithfulness and answer relevancy; integrated LangSmith for end-to-end chain tracing and cost monitoring
- Built scalable ingestion with **semantic chunking** (with overlap), vector embeddings in **Qdrant Cloud**, and multiple query modes (standard + search-only)
- Containerized with **Docker** (non-root, health checks), deployed on **AWS** with **GitHub Actions CI/CD**, 70% test coverage via **pytest**

Multi-Source RAG + Text-to-SQL System

- Designed a dual-mode AI query platform with intelligent routing across Document RAG and **Text-to-SQL pipelines**, deployed serverlessly on **AWS Lambda** with full observability.
- Achieved a **40-60% reduction** in **OpenAI API** costs via SHA-256 content-based chunk and **embedding deduplication**.
- Used **Docling HybridChunker** for structure-preserving document parsing, improving downstream **RAGAS** faithfulness and relevancy scores
- Built **Text-to-SQL** pipeline with natural language to **SQL** conversion, human-in-the-loop approval workflow, and structured error responses
- Deployed on **ARM64 Lambda** with end-to-end monitoring via **OPIK tracking** and **CloudWatch**; managed cross-platform **Docker** builds

Master's Research Project - Cyber Security Approaches for IoT based on ML and Cloud

- Implemented, Compared and deployed **ML models** for cybersecurity attack detection in the IOT domain using **MongoDB** for data ingestion and storage
- Designed and implemented a scalable cloud-based architecture using **Flask** and **docker containers**, deploying the application on **Azure Kubernetes Service (AKS)**
- Automated cloud resource provisioning using **Terraform**

- Implemented CI/CD pipelines using **GitHub Actions** incorporating automated security checks, including SAST and dependency audits.
- Utilized **Prometheus** and **Grafana** for monitoring of the AKS cluster, validating robust network performance and stable resource utilization

EXPERIENCE

09/2023 – 11/2023 **Working student (Fixed Term Contract) | Truemetrics | Germany**

- Workflow Automation: Engineered automated video processing and data management pipelines using **Python** to accelerate large-scale dataset preparation for **computer vision** systems.
- Spatial Data Annotation: Extracted and mapped precise coordinates from Google Maps to identify navigational features (entrances and parking positions), enhancing location-based modeling.
- Quality Assurance: Implemented rigorous QA protocols for computer vision **dataset validation**, achieving and maintaining **95% annotation accuracy** and consistency.

09/2022 – 11/2022 **Computer Vision Intern (Mandatory Internship) | Bosch Research | Germany**

- End-to-End Perception Pipeline: Designed and trained a **deep learning**-based object classifier in **PyTorch**, integrating it into a scalable autonomous driving pipeline using the **Audi A2D2** dataset.
- Optimized Data Ingestion: Engineered two high-performance **Python DataLoaders** for synchronized multi-sensor inputs and calibration metadata, significantly reducing data preparation overhead and enabling reproducible training runs.
- System Architecture & Refactoring: Refactored the codebase using **object-oriented design (OOP)**, introducing a shared base class for detection and classification workflows that boosted **system efficiency by 80%** and improved long-term maintainability.

07/2019 – 08/2019 **Machine Learning Intern (Mandatory internship) | Verzeo | India**

- Model Deployment: Trained and deployed machine learning classification and regression models using **AWS SageMaker**.
- Data Engineering (ETL): Extracted, transformed, and loaded data from multiple sources via **APIs** and web scraping to engineer features for a robust recommender system.
- Pipeline Optimization: Designed training and validation pipelines involving rigorous data cleaning and **Exploratory Data Analysis (EDA)** to ensure high model accuracy. Implemented training and validation pipelines for **classification** and **regression algorithms**, ensuring performance and accuracy.

EDUCATION

01/2022 – 07/2027 **Technical University Ilmenau | Ilmenau | Germany**

Master of Science in Research in Computer and System Engineering

08/2016 – 08/2020 **Visvesvaraya Technological University | Bengaluru | India**

Bachelor of Engineering in Computer Science

LANGUAGES

German: Beginner (A2) | English: Fluent (Professional Proficiency)

CERTIFICATIONS

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- Microsoft Certified: Azure AI Fundamentals (AI-900)
 - Machine Learning Statistical Foundations Professional Certificate by Wolfram Research
 - Vector Databases Professional Certificate by Weaviate
 - Agentic AI & Retrieval Augmented Generation Engineering 6 months program – ONGOING